Metal forming

Metal forming : making desired shape of comp by generating plastic def in material with mechanical loads.

$\boxed{\sigma_y < \sigma < \sigma_{ut}}$ with no wastage of material

Yield stress — metal forming stress — ult stress

Cold working	Hot Working
• Working below recrystallization temp	• above recrystalization temp (R.C)
• strength & Hardness ↑↑ b/c of strain hardening effect	• ductility & toughness ↑↑ b/c grain flow without dislocation
• Grain structure is changed	
• scales and oxides not formed	• Scales and oxides formed (out side temp ... so billet temp ... black colour ... form ...)
surf finish is good	• surf finish is rough
• coeff friction (μ) ↓ ie 0.1 to 0.2	• μ ↑ → 0.4 to 0.6
• force req ↑ • dimensions accuracy ↑	• Force req ↓, • dimensions accuracy ↓

Warm working: heated just below Recrystallization temp (R.C) to get adv of both Cold and hot working

Isothermal working: working at const temp by heating rollers also

Recrystalization temp (RCT): It is *Exact* min temp at which new crystals formation completed

Recovery 400°c — 500°c Recrystallization — 700 (recrystalization temp) — 850°c Grain Growth

voids (or) holes (or) defects

Dislocation & point defect

dislocation & defect vanishinshed (def repaired)

New crystal fromation started means new grains

New crystal formation Completed before recrystalization temp

grain size increased

Recrystalization temp(RCT) after Cold working: RCT is less than normal comp for Cold worked comp due to residual stresses and internal energy stored in mat less heat required

RCT after hot working: RCT is same before and after Hot working due to no internal energy and residual stresses

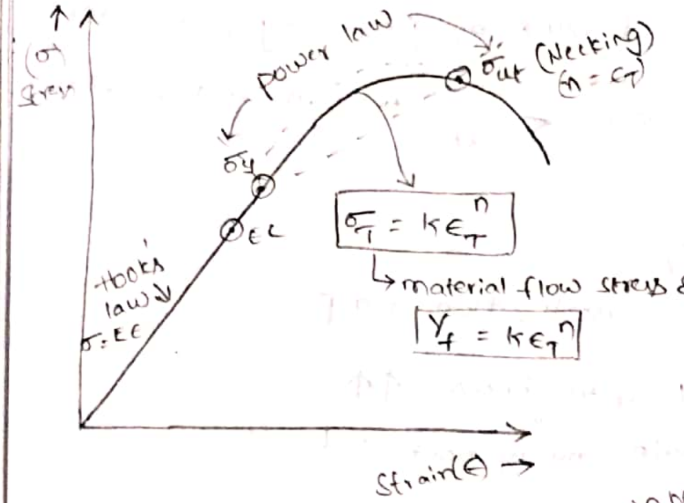
Power law of strain hardening: True stress = $\frac{\text{Load}}{\text{Inst area}}$

Y_f = flow stress (σ) True stress

k: Strain hardening coeff

ϵ_T = True strain

n = Strain hardening exponent $(0<n<1)$



$\sigma_T = \sigma_E(1+\epsilon_E)$

$\sigma_T = \sigma_E \times \left(\frac{A_i}{A_f}\right)$

$= \sigma_E \times \left(\frac{d_i}{d_f}\right)^2$

$\sigma_T = \sigma_E \times \frac{l_f}{l_i}$

ϵ_T = True strain

$\epsilon_T = \ln\left(\frac{A_i}{A_f}\right)$

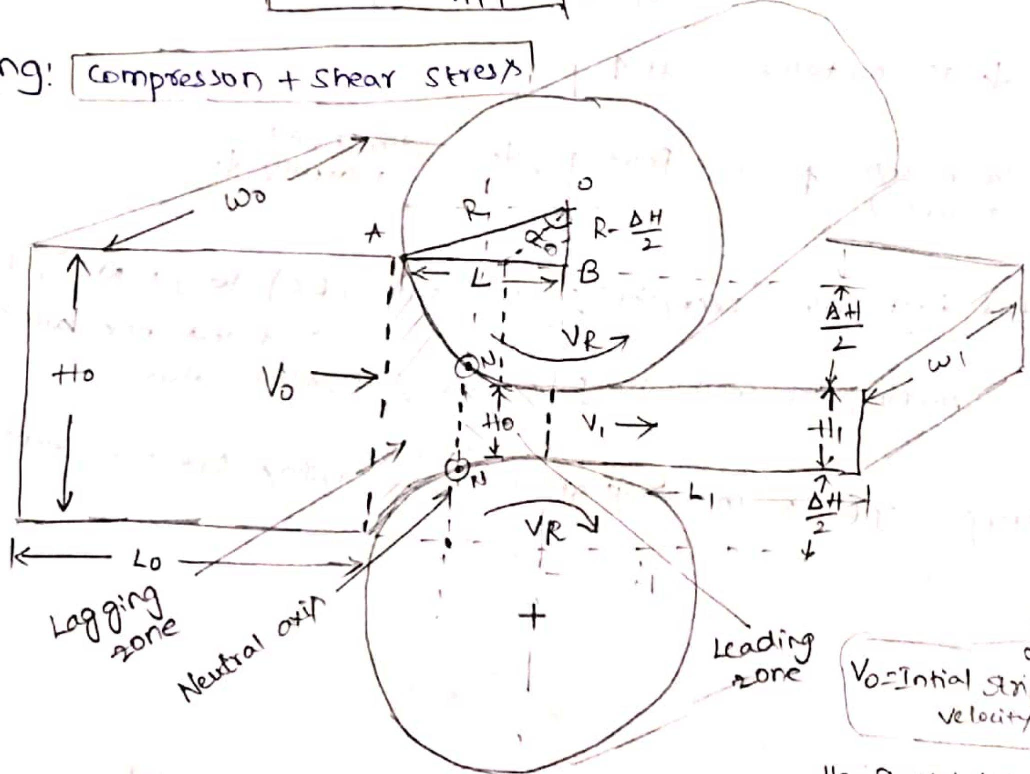
$= 2\ln\left(\frac{d_i}{d_f}\right)$

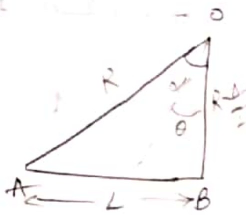
$\epsilon_T = \ln\left(\frac{l_f}{l_i}\right)$ $\qquad \epsilon_T = \ln(1+\epsilon_E)$

strain hardening coeff, strain hardening exponent

Avg flow stress *Imp* $\boxed{(Y_f) = \frac{k\epsilon_T^n}{n+1}}$ $\qquad$ Flow stress $\boxed{\sigma_T = k\epsilon_T^n}$

I Rolling: $\boxed{\text{Compression + shear stress}}$





L = Length of deformation

R = Radius of roller

V_R - Roller velocity

V_S = Strip velocity

N = Roller speed (rpm)

α = def Angle (or) bite angle

V_0 = Intial strip velocity $\quad$ V_1 = Final strip velocity

H_θ = Thickness of strip at an angle of θ from exit

H_0 = Intial thickness $\quad$ H_1 = Final thickness

W_0 = Intial width $\quad$ W_1 = Final width

L_0 = Inital length of strip $\quad$ L_1 = final length

- No change in Volume *Imp* $\boxed{H_0 W_0 L_0 = H_1 W_1 L_1}$

- Elongation factor $= \frac{L_1}{L_0} = \frac{A_0}{A_1} = \frac{H_0 W_0}{H_1 W_1}$

α = Angle subtended by def zone at center of roll

- Volume flow rate discharge = Area x Velocity = C

$A \times V = C$

$A_0 V_0 = A_1 V_1$

$\boxed{H_0 W_0 V_0 = H_1 W_1 V_1}$ *Imp*

V_0 = Intial strip velocity

V_1 = final " "

→ change in thickness *Imp $\Delta H = H_0 - H_1$

$\Delta H = D(1-\cos\alpha)$

roller dia

roller radius

bite angle

$\tan\alpha = \sqrt{\frac{\Delta H}{R}}$ *Imp* use this formula to find α

→ Length of deformation $L = \sqrt{R(\Delta H)}$

→ change in length at angle θ from exit $H_0 - H_1 = R\theta^2$ radians

→ Backward (or) lagging zone

$V_S < V_R$

Slip $= \frac{V_S - V_R}{V_R} = -ve \Rightarrow \frac{V_0 - V_R}{V_R} \times 100$ entry (or) initial strip velocity

pressure ↑ up to Neutral plane

→ Forward (or) leading zone

$V_S > V_R$

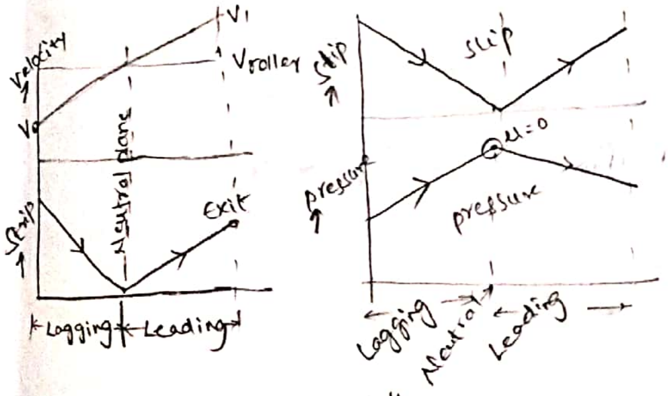
Slip $= \frac{V_S - V_R}{V_R} = +ve = \frac{V_1 - V_R}{V_R} \times 100$ exit strip velocity

pressure ↓ from Neutral plane

→ Neutral plane $(V_S = V_R) = \frac{\pi D N}{60}$

Slip = 0

Pressure is max at Neutral axis



Rolling Analysis friction coeff $\mu \geq \tan\alpha$ *Imp* condition for rolling

$\mu = \tan\beta$

$\tan\beta \geq \tan\alpha$

friction angle $\boxed{\beta \geq \alpha}$ deformation angle

Imp

→ Max reduction possible per pass in rolling *Imp* $\Delta H = \boxed{H_0 - H_1 = \mu^2 R}$

$\Delta H \uparrow \propto \alpha \uparrow$

$\Delta H \propto \mu$

$\Delta H \propto R$

In this condition α = β — friction angle, deform angle

$\alpha = \beta$, $\mu = \tan\beta$

No. of passes required in rolling operation

Case-① Const draft (red in thickness)

draft $\Delta H = H_0 - H_1 = H_1 - H_2$

Min No. of passes $\boxed{n_{min} = \frac{H_0 - H_f}{\mu^2 R}}$ max possible ΔH

Case ② Elongation factor (k) const

$k = \frac{L_1}{L_0} = \frac{A_0}{A_1} \Rightarrow \frac{L_1}{L_0} = \frac{L_2}{L_1}$ (or) $\frac{A_0}{A_1} = \frac{A_1}{A_2}$

$\boxed{\frac{A_0}{A_n} = k^n = \frac{L_1}{L_0}}$

Imp

Case ③ Const reduction in each pass

reduction (r) $= \frac{H_0 - H_1}{H_0} = \frac{H_1 - H_2}{H_1}$

$\boxed{H_n = (1-r)^n H_0}$ Imp

reduction value (Ex 0.2, 0.3)

Plane strain condition $W_0 = W_1$ (or) deformation

In problems not given two width take $W_0 = W_1$

Imp

Scanned with CamScanner

Rolling pressure (P_{avg})

Case-I: The avg roll pressure is given (P_{avg})

Rolling force $= F_R = P_{avg} \times A_p$

$$\boxed{F_R = P_{avg} \times L \times W}$$

$$F_R = P_{avg} \times \sqrt{R\Delta H} \times W$$

Imp*

Case-2 Ideal rolling ($\mu = 0$)

condition (A). The strip material is rigid and plastic plain strain condition. Width (W) const

$$P_{avg} = \sigma_1 - \sigma_2 = \frac{2}{\sqrt{3}} \sigma_y$$

$$\boxed{F_R = \frac{2}{\sqrt{3}} \sigma_y . L . W}$$

Condition B: Strip material is rigid and strain hardened

Avg flow stress $\boxed{\bar{Y}_f = \frac{K\epsilon_1^n}{n+1}}$

Rolling force $= \boxed{F_R = \frac{2}{\sqrt{3}} \bar{Y}_f . L . W}$

Case-3: Considering friction b/w roll and work piece ($\mu \neq 0$) (This for

condition (A): The strip material is rigid and plastic (σ_y)

put $\mu = 0$ to get Case-2 formulas

$$\boxed{F_R = \frac{2}{\sqrt{3}} \sigma_y \left(1 + \frac{\mu L}{2 H_{avg}}\right) \times L \times W}$$

Condition (B) The strip material is rigid and strain hardened ($\bar{Y}_f$)

Imp

$$\boxed{F_R = \frac{2}{\sqrt{3}} \bar{Y}_f \left(1 + \frac{\mu L}{2 H_{avg}}\right) \times L \times W}$$

$$\boxed{H_{avg} = \frac{H_0 + H_1}{2}}$$

Power required in rolling Total power $= \boxed{P_{total} = 2 \text{ power}_{\text{per roller}}}$

$\text{power}_{\text{per roller}}$ = Torque $\times$ angular speed

$= T \times \omega$

$$\omega = \frac{V_1/60}{R} \text{ rad/s}$$

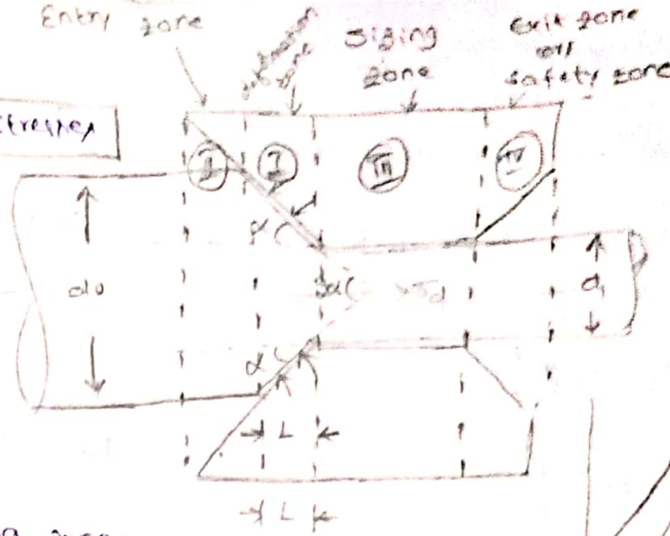
deformation length

Imp

$$\boxed{\text{Power}_{\text{per roller}} = F_R \times \frac{L}{2} \times \frac{2\pi N}{60}}$$

Wire drawing

Tensile + compression + shear stresses



2α = full die angle

α = semi/half die angle

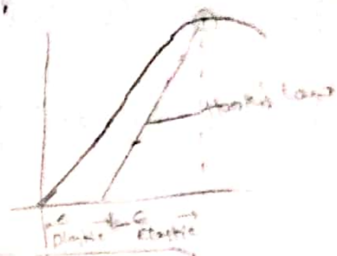
L = length of deformation

II : Entry zone (or) lubricating zone :

I : Deformation zone

III : Sizing zone → converting elastic deformation into plastic deformation

IV : exit (or) safety zone → collect lubricants ~~tards~~ to avoid formation of surface cracks



$\epsilon_{Total} = \epsilon_{plastic} + \epsilon_{elastic}$

I Deformation zone

Case I : Const length deformation

reduction in cross section area increase by increasing die angle but due to sudden change in c/s area more drawing stress req and some time wire will break in def zone

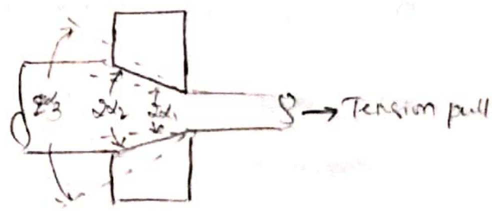
So large die angle are not preferable

Case-II $\alpha_3 < \alpha_1 < \alpha_2$ Const red in c/s area

The length of def zone Increase by reduction die angle

II Entry (or) lubricating zone

Lubricant	Colour	Effect
Gaseous lubricant	Black/brunt	$\mu\uparrow$ Heat $\uparrow$ burring
Liq lubri	metallic/dull	$\mu\downarrow$ Heat $\downarrow$
Solid powder lubricant	Glittery (or) silver	microscratches on surfaces



Preparation of Work

1. Annealing : heat and cold the wire in furnace
2. Cleaning
   - i, Acid pickling : Removing chemicals, corrosion from surf of wire
   - ii, Alkaline cleaning : Removing oil, greese from surf of wire
   - iii, phosphate cleaning : To avoid direct cont with die & wire
   - iv, Lubrication : To reduce friction to dissipate heat
3. Pointing : making taper at end

Drawing stress

Imp*

Case ①: Ideal wire drawing ($\mu = 0$)

condition Ⓐ: The wire metal is rigid & plastic

$\frac{F_d}{A_1} = \sigma_d = \sigma_y \times \ln\left(\frac{A_0}{A_1}\right)$

drawing stress, yeild stress

$\boxed{F_d = \sigma_y \times A_1 \times \ln\left(\frac{A_0}{A_1}\right)}$

final area

Condition Ⓑ: Wire metal is rigid & Strain harderd

$\sigma_d = \bar{Y}_f \times \ln\left(\frac{A_0}{A_1}\right)$

$\boxed{F_d = \bar{Y}_f \times A_1 \times \ln\left(\frac{A_0}{A_1}\right)}$

Case ②: Actual wire drawing ($\mu \neq 0$) → No need to remember formula given in gate

Frictional factor $\boxed{B = \frac{\mu}{\tan\alpha} = \mu \cot\alpha}$

cond Ⓐ: Wire is rigid & plastic

$\sigma_d = \sigma_y \times \left(\frac{1+B}{B}\right)\left(1 - \left(\frac{A_1}{A_0}\right)^B\right)$

$\boxed{F_d = \sigma_y \times A_1 \times \left(\frac{1+B}{B}\right)\left[1 - \left(\frac{A_1}{A_0}\right)^B\right]}$

Cond Ⓑ: Wire is rigid and strain hardend

$\sigma_d = \bar{Y}_f \cdot \left(\frac{1+B}{B}\right)\left[1 - \left(\frac{A_1}{A_0}\right)^B\right]$

$\boxed{F_d = \bar{Y}_f \times A_1 \times \left[\frac{1+B}{B}\right]\left[1 - \left(\frac{A_1}{A_0}\right)^B\right]}$

Total drawing stress (Incld die cond) $\boxed{\sigma_{Total} = \sigma_y + (\sigma_d - \sigma_y)\, e^{-\frac{2\mu L}{R_1}}}$ not imp for gate

Max ~~per~~centage reduction $\boxed{\%\ \text{Reduction} = \frac{A_0 - A_1}{A_1} \times 100}$

Imp*

Case ①: max percentange reduction ideal wire drawing ($\mu = 0$)

Cond Ⓐ: For perfect plastic material

$\boxed{\text{drawing stress} = \text{yeild stress}}$ ← condition for max possible reduction

$\sigma_d = \sigma_y$

$\sigma_d = \sigma_y \ln\left(\frac{A_0}{A_1}\right) = \sigma_y$

$\ln\left(\frac{A_0}{A_1}\right) = 1$

$\frac{A_0}{A_1} = e^{1}$

Max. % reduction $= \frac{A_0 - A_1}{A_0} \times 100 \Rightarrow \left(1 - \frac{A_1}{A_0}\right) \times 100$

$\Rightarrow (1 - e^{-1}) \times 100 \Rightarrow$ 63.2% Imp*

Cond Ⓑ: For strain hardened material

$\sigma_d = Y_f$

$\cancel{Y_f} \ln \frac{A_0}{A_1} = \cancel{Y_f}$

$\frac{K\epsilon_1^n}{n+1} \ln \frac{A_0}{A_1} = \cancel{K}\epsilon_1^n$

$\ln\left(\frac{A_0}{A_1}\right) = n+1 \Rightarrow \frac{A_0}{A_1} = e^{n+1}$

Max red $= \frac{A_0 - A_1}{A_0} \times 100$

Case ②: max percentage red per pass in actual wire drawing ($\mu \neq 0$)

Cond Ⓐ: rigid and plastic

$\sigma_d = \sigma_y$

$\sigma_y \left(\frac{1+B}{B}\right)\left[1-\left(\frac{A_1}{A_0}\right)^B\right] = \sigma_y$

$1-\left(\frac{A_1}{A_0}\right)^B = \frac{B}{1+B}$

$\left(\frac{A_1}{A_0}\right) = \left(\frac{1}{1+B}\right)^{1/B}$

Max % red $= \left(1-\frac{A_1}{A_0}\right)\times 100 = \left[1-\left(\frac{1}{1+B}\right)^{\frac{1}{B}}\right]\times 100$

Cond (B) rigid and strain hardend

$\sigma_d = \sigma_y$

$Y_f \left(\frac{1+B}{B}\right)\left[1-\left(\frac{A_1}{A_0}\right)^B\right] = Y_f$

$\frac{K\epsilon_1^n}{n+1}\left(\frac{1+B}{B}\right)\left[1-\left(\frac{A_1}{A_0}\right)^B\right] = K\epsilon_1^n$

$\frac{A_1}{A_0} = \left[1-(n+1)\left(\frac{B}{1+B}\right)\right]^{1/B}$

Max % red $= \left(1-\frac{A_1}{A_0}\right)\times 100$

$\Rightarrow \left[1-\left[1 - (n-1)\left(\frac{B}{1+B}\right)\right]^{1/B}\right]$

Imp

$\boxed{\text{Max possible \% reduction in area per pass during an ideal plastic material without friction is of order of 63.2\%.}}$

Tube drawing (In gate not asking on stresses in tube drawing from this tube drawing)

$\boxed{\text{power req in tube drawing} \quad P = F_{tube} \times V_{exit}}$ → tube exit velocity

Case - ① Ideal tube drawing ($\mu = 0$)

Cond Ⓐ: The tube is rigid and plastic

$\sigma_d = \sigma_y \ln\left(\frac{h_0}{h_1}\right)$

$F_d = \sigma_y \times A_1 \times \ln\left(\frac{h_0}{h_1}\right)$

Cond B material is rigid and strain hardend

$\sigma_d = \bar{Y}_f \ln\left(\frac{h_0}{h_1}\right)$

$F_d = \bar{Y}_f \cdot A_1 \cdot \ln\left(\frac{h_0}{h_1}\right)$

Case ②: Actual tube drawing ($\mu_1 \neq \mu_2 \neq 0$)

friction factor $B = \frac{\mu_1 + \mu_2}{\tan\alpha - \tan\beta}$

Cond Ⓐ material rigid and plastic

$\sigma_d = \sigma_y \left(\frac{1+B}{B}\right)\left[1-\left(\frac{h_1}{h_0}\right)^B\right]$

$F_d = \sigma_y \times A_1 \times \left(\frac{1+B}{B}\right)\left[1-\left(\frac{h_1}{h_0}\right)^B\right]$

Cond B: rigid and strain hardend

$\sigma_d = \bar{Y}_f \left(\frac{1+B}{B}\right)\left(1-\left(\frac{h_1}{h_0}\right)^B\right)$

$F_d = \bar{Y}_f \times A_1 \times \left(\frac{1+B}{B}\right)\left[1-\left(\frac{h_1}{h_0}\right)^B\right]$

Case ③ movable mandrel $\mu_1 = \mu_2 = \mu$

$B = \frac{\mu_1 - \mu_2}{\tan\alpha} = 0$

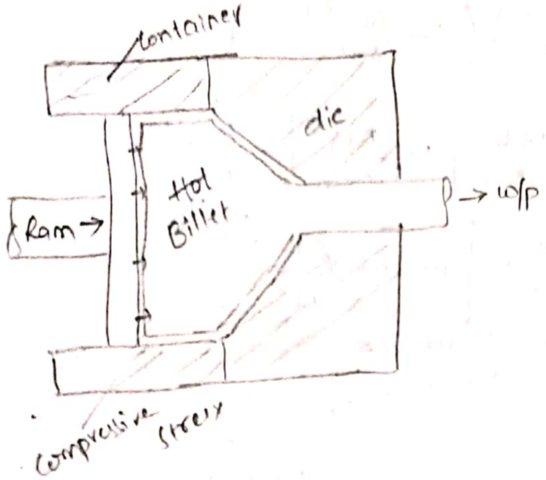
$\sigma_d = \sigma_y \left(\frac{1+B}{B}\right)\left[1-\left(\frac{h_1}{h_0}\right)^B\right]$

L-Hospital rule $\sigma_d = \sigma_y \ln\left(\frac{h_0}{h_1}\right)$ or $\bar{Y}_f \ln\left(\frac{h_0}{h_1}\right)$

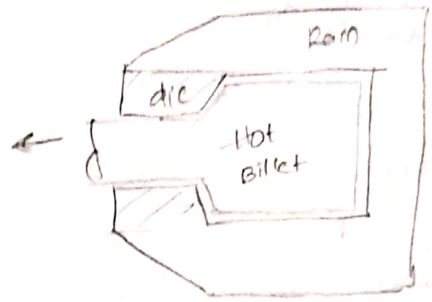
4. Extrusion :

Extrusion is more efficient than wire drawing for reduction in c/s area

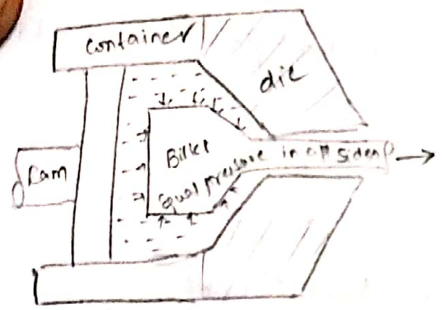
Forward (or) direct extrusion



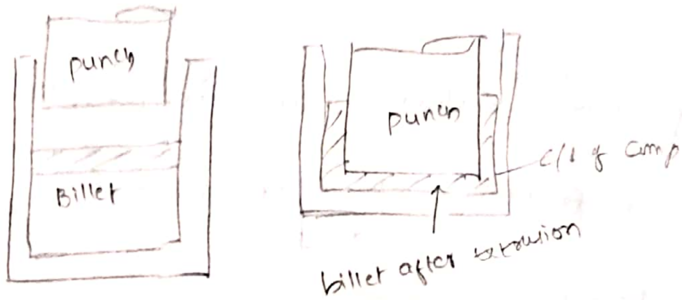
Backward (or) Indirect extrusion



Hydrostatic extrusion



Impact extrusion



Ram pressure and Extrusion load

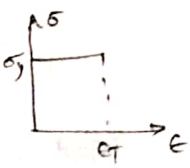
$$\boxed{\text{power req in extrusion} = F_E \times V_{ram}}$$

Imp* case-① Ideal extrusion $(\mu = 0)$

Cond Ⓐ : Billet is rigid and plastic

Extrusion stress $\boxed{\sigma_E = \frac{F_d}{A_0}} = \sigma_y \times \ln\left(\frac{A_0}{A_1}\right)$ Initial Area, extrusion ratio

$F_d = \sigma_y \times A_0 \times \ln\left(\frac{A_0}{A_1}\right)$ Initial area

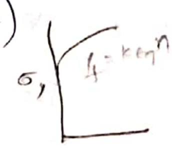


Cond B Billet is rigid and strainhardened

$Y_f = \frac{K \epsilon^n}{n+1}$ avg flow stress, not case consider avg flow stress m'y

Ram pressure $p = \bar{Y}_f \cdot \ln\left(\frac{A_0}{A_1}\right)$

$F_E = \bar{Y}_f \cdot A_0 \times \ln\left(\frac{A_0}{A_1}\right)$



Case-2 Indirect extrusion $(\mu_{container} = 0, \mu_{die} \neq 0)$

$B = \mu \cot \alpha$

Ram pressure $P_{ram} = \sigma_y \left(\frac{1+B}{B}\right)\left[1 - \left(\frac{A_1}{A_0}\right)^B\right]$

Case ③ not at all Imp

$F_E = \sigma_y \times A_0 \times \left(\frac{1+B}{B}\right)\left(1 - \left(\frac{A_1}{A_0}\right)^B\right)$

Case ③ Direct extrusion ($\mu_{cont} \neq 0$, $\mu_{die} \neq 0$)

Ram pressure $P_r = \bar{\sigma}_y \left(a + b\ln\left(\frac{A_0}{A_1}\right) + \frac{2L_0}{d_0}\right)$

Case ③ formula given in gate exam

extrusion force $F_E = \bar{\sigma}_y \times A_0 \times \left(a + b\ln\left(\frac{A_0}{A_1}\right)\right)$

$F_E = \bar{Y}_f \times A_0 \times \left[a + b\ln\left(\frac{A_0}{A_1}\right)\right]$

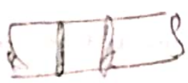
Ideal Case - Not given formula
Actual Case : Formula given

Imp
Case ④ Hot extrusion

Ram pressure $P_{ram} = K \ln\left(\frac{A_0}{A_1}\right)$

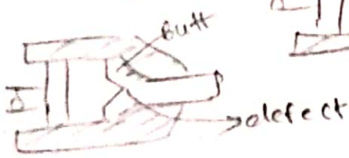
Extrusion force $F_E = K \cdot A_0 \times \ln\left(\frac{A_0}{A_1}\right)$ — extrusion const (mpa) at temp t

Defects in extrusion

1. Bamboo defect  - formed due to non-uniform velocity

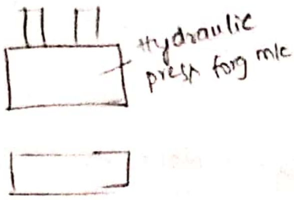
Imp 2. Central burst (or) pin hole defect 

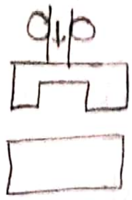
due μ b/w die and billet Central cracks are generated

3. Fish tail defect 

5. Forging

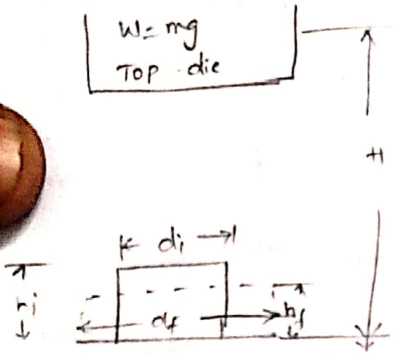
press forging   drop hammer forging





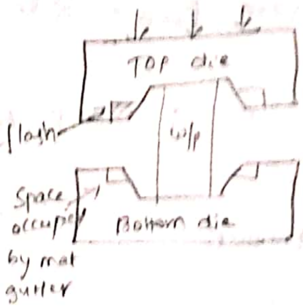
forging force applied gradually   Sudden forging force

Drop hammer forging   closed die forging



$$F_{avg} = \frac{W \cdot H}{h_i - h_f}$$

$$F_{avg} = \frac{(mg + PA) H}{h_i - h_f}$$



Imp
Gutter: free space provided in die to occupy excess volume of raw material

Flash: excess Volume of material occupied in gutter

Analysis of forging (open die forging only & closed die not in syllabus)

$\boxed{\text{Volumetric Change} = 0 \quad A_o h_o = A_1 h_1}$

$$\frac{\pi}{4} d_o^2 h_o = \frac{\pi}{4} d_1^2 h_1$$

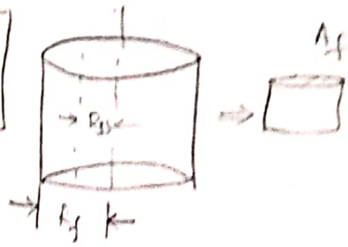
$$\boxed{d_o^2 h_o = d_1^2 h_1}$$

Forging force on cylindrical body

Graph

$$\boxed{R_{ss} = R_f - \frac{h_f}{2\mu} \ln\left(\frac{1}{\sqrt{3}\,\mu}\right)}$$

radius   Final height



Case ① Ideal condition ($\mu = 0$)

Cond Ⓐ : w/p rigid and plastic

$$F_{max} = \sigma_y \times A_f$$

Cond Ⓑ : w/p rigid and strain hardened

$$F = Y_f \cdot A_f = (K \epsilon_T^n) A_f$$

Remember this two formulas
In forging it is enough

Case ② Actual forging ($\mu \neq 0$)

Cond A : w/p is rigid and plastic

$$F = \sigma_y \times A_f$$

Cond Ⓑ w/p rigid and Strain hardened

*V.Imp

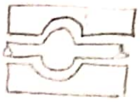
$$\boxed{F = Y_f \cdot A_f \left(1 + \frac{2\mu r_f}{3 h_f}\right)}$$

forging operations

Fullering | Drawing | upsetting | Edgering | Blocking

die

die

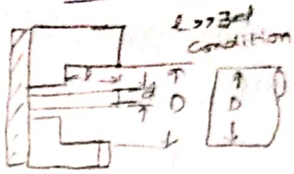


Blocking Impression die forging

Imp

closed upsetting

$l >> 3d$ condition



Imp $\boxed{D = 1.5d}$

Problems Forging

- Energy = force × distance → change in Height of w/p
- open die forging → Semi closed die → closed die → Trimming (or) cutting
- connecting rod

Fullering - edging - blocking - Finishing - trimming

Rolling problems

1. $\tan\alpha = \sqrt{\frac{\Delta H}{R}}$ when we substitue and get value of α in degree only

   or $\alpha = \tan^{-1}\sqrt{\frac{\Delta H}{R}}$ → degree only

   But when we substitue we should substitue in rad only.

2. $\tau_u = 0.6\,\sigma_u$

   $\sigma_y = 0.5\,\sigma_u$

3. $\sigma = 1400\,\epsilon^{0.33}$ → max load $\epsilon = 0.33 = \frac{1}{3}$

Extrusion

Tooth post tubes are produced by Impact extrusion

- Extrusion ratio $= \frac{A_0}{A_1}$

Extrusion : long continious metal

Forging : difficult parts (connecting rod, crank shaft)

metal spinning: spinning (or) metal turning axisymmetry (House Hold large వస్తువు)

Explosive welding: Bounding sheet of corrosive resist to base metal metal

cladding operation

Sheet Metal

plate thickness >> 5mm
Sheet = thickness < 5mm

Punching

Punch size = Hole size (H.S)

die size = H.S + clearance

$$\boxed{D.S = P.S + 2C_r}$$

Blanking

Die size (D.S) = Blank size

Punch size (P.S) = D.S − clearance

$$\boxed{P.S = D.S - 2C_r}$$

Max punch force F_{max} = Ultimate Shear strength × Shear Area.

$F_{max} = \tau_u \times \pi d t$ (d)

$= \tau_u \times 4at$ (square, a)

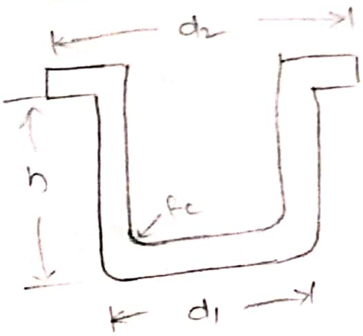
$= \tau_u = 2(a+b)t$ (rectangle, a × b)

min punch dia $= \sigma_{Ac} = \dfrac{F_{max}}{\frac{\pi}{4}d^2}$

$$\boxed{d_{min} = \frac{4t\tau_u}{\sigma_{Ac}}}$$

Enererg required $= E = F_{max} \times$ Penatration

Flange Cup



1^{st} draw = PRR $= \dfrac{D_b - d_1}{D_b}$

2^{nd} draw = DRR $= \dfrac{d_1 - d_2}{d_1}$

3^{rd} draw = DRR $= \dfrac{d_2 - d_3}{d_2}$

$\frac{\pi}{4}D_b^2 = \frac{\pi}{4}d_1^2 + \pi d_1 h + \frac{\pi}{4}(d_2^2 - d_1^2)$

$$\boxed{D_b = \sqrt{d_2^2 + 4d_1 h}}$$

Ex $d_3 = 10.13 < 12$

$$\boxed{\text{drawing load \& force} = F_d = \pi D_b t \sigma_{ut}}$$

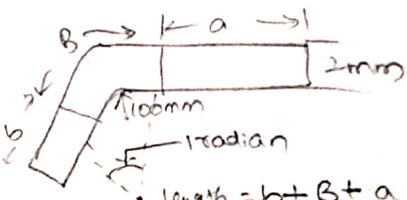
Ex:

Bend allowance $= (R + kt)\,\alpha_b$ radians

(Bend radius, Stretch factor 1/2, thickness)

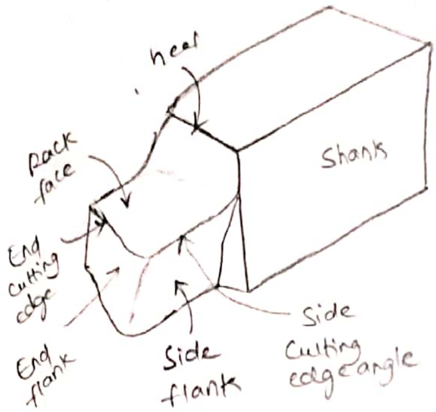
$= \left[100 + \left(\frac{1}{2} \times 2\right)\right] \times 1$

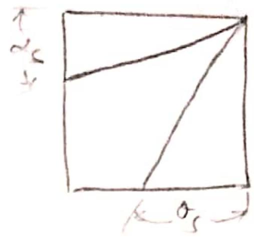
$= 101$ mm



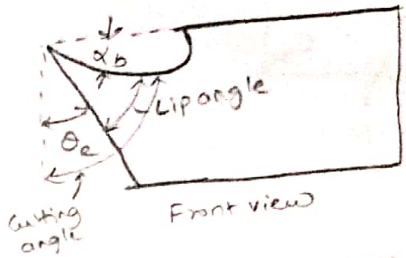
Length = b + B + a

Metal Cutting

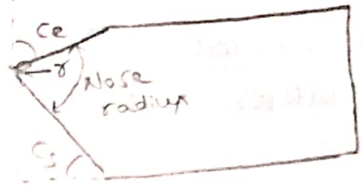




Side view



Front view



Top view

American Standard association system. ASA

V.Imp

$$\boxed{\alpha_b - \alpha_s - \theta_e - \theta_s - C_e - C_s - \gamma}$$

α_b = Back rack angle

α_s = side rack angle

θ_e = end relif angle (or) clearance

θ_s = side relief angle

Ce = end cutting edge angle

Cs = Side cutting edge angle

γ = Nose radius

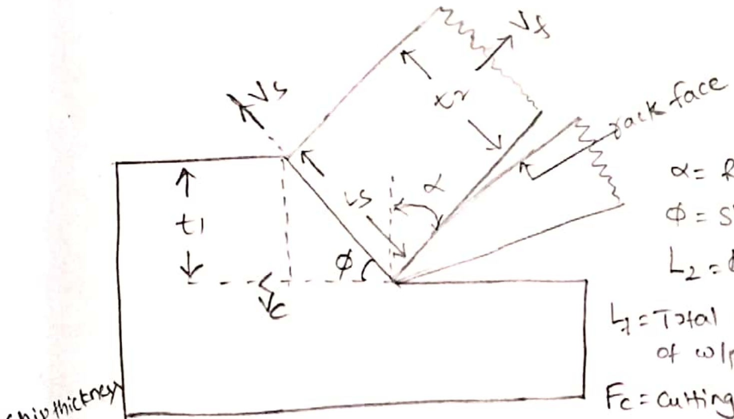
Orthogonal rack system ORS

$$\boxed{i - \alpha - \theta_s' - \theta_e' - C_e - \lambda - \gamma}$$

$$\boxed{\lambda = 90 - C_s}$$ *V.Imp

$\tan i = \sin\lambda \cdot \tan\alpha_b - \cos\lambda \tan\alpha_s$

$\tan\alpha = \cos\lambda \cdot \tan\alpha_b - \sin\lambda \tan\alpha_s$



α = Rank angle

ϕ = Shear angle

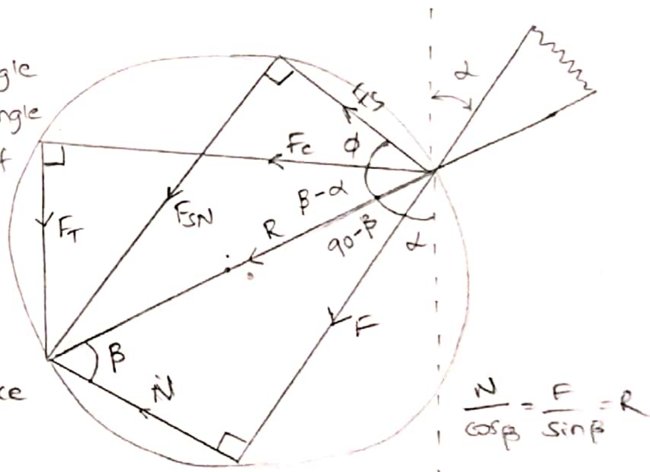
L_2 = Length of chip

L_1 = Total length of w/p = πD

F_c = Cutting force

F_T = Thrust force

F_s = Shear force



Chip thickness ratio

$$\boxed{r = \frac{t_1}{t_2} = \frac{L_2}{L_1} = \frac{V_f}{V_c}}$$ V.Imp — cutting velocity — V.Imp

$$\frac{N}{\cos\beta} = \frac{F}{\sin\beta} = R$$

$$\boxed{\tan\phi = \frac{r\cos\alpha}{1 - r\sin\alpha}}$$

$$\frac{V_c}{\cos(\phi-\alpha)} = \frac{V_s}{\cos\alpha} = \frac{V_f}{\sin\phi}$$

V.Imp

$$R = \boxed{\frac{F_c}{\cos(\beta-\alpha)} = \frac{F_s}{\cos(\phi+\beta-\alpha)}} = \frac{F_T}{\sin(\beta-\alpha)} = \frac{F_{SN}}{\sin(\beta-\alpha+\phi)}$$

V Imp

$$\frac{F_T + F_c\tan\alpha}{F_c - F_T\tan\alpha} = \boxed{\tan\beta = \mu}$$ Imp

$$\boxed{r = \frac{V_f}{V_c} = \frac{\sin\phi}{\cos(\phi-\alpha)}}$$ *V.Imp

V.Imp

$$\boxed{\text{Power} = F_c \times V_c}$$ m/s $\quad V_c = \frac{m}{min} = \frac{V_c}{60}$ (m/s)

Shearing Take place at min energy condition $= \boxed{2\phi + \beta - \alpha = 90}$

Marchant Const (or) Machine Const $\quad C_m = 2\phi + \beta - \alpha$

Tool life

V.Imp*

$$\boxed{V_c T^n = C}$$

Cutting speed (V_c) $\left(\frac{m}{min}\right)$

Tool life (min)

Taloy's Constant $\left(\frac{m}{min}\right)$

$$V f^p d^q T^n = C$$

feed

depth of cut

Optimal Cutting Speed

$$V_{opt} = \frac{C}{\left[\left(\frac{1}{n} - 1\right)\left(\frac{C_e}{C_m} + T_c\right)\right]}$$

Taloy's Const

m/c running cost

0 if not given

depreciation of tool regrind cost

Material removal rate

feed (mm/rev)

depth of cut (mm)

cutting velocity $\left(\frac{m}{min}\right)$

$$\boxed{(MRR) = F\,D\,V}$$

If F and d given

$t_1 = f$

$b = d$

→ V.Imp

If d and W given

$t_1 = d$

$b = W$

If f and W given

$t_1 = f$

$b = W$

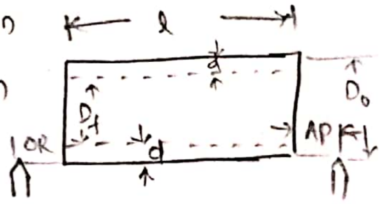
Scanned with CamScanner

Machining and Machine tools

Lath

$$\boxed{\text{Time/Cut} = \frac{L}{F \times N}} \ \text{min}$$

(units noted: mm, mm/rev × rev/min; F: mm/rev; N: rpm)



cutting velocity $V_c = \dfrac{\pi D N}{1000}$ mm/min

$$\text{No. of cuts} = \frac{D_o - D_f}{2d}$$

Drilling

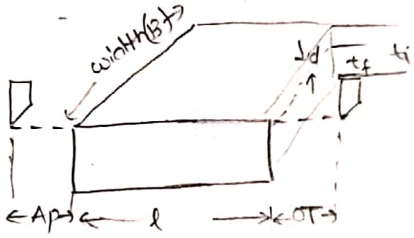
0.5D → dia of drill hole / difference

L = CAP + AP + t + OR

$$\boxed{\text{Time/hole} = \frac{L}{F \times N}}$$

Shaper

$$\boxed{\text{Time/cut} = \frac{B}{F} \times \frac{L}{V_c}(1+M) = \frac{B}{F} \times \frac{1}{N}}$$



$$\boxed{V_c = LN(1+M)}$$

$$\boxed{M = \frac{V_c}{V_r} = \frac{T_r}{T_c}}$$

T_r = return time, T_c = cutting time

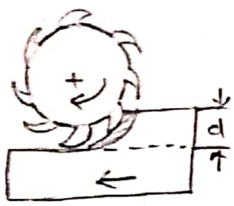
L = AP + l + OT

L = stroke length

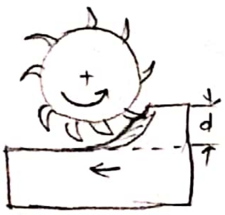
$$\text{No. of Cuts} = \frac{t_i - t_f}{d}$$

Total machining time = Time/cut × No. of cuts

milling



Down milling



up milling

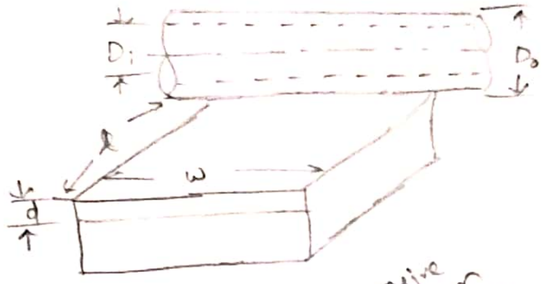
Grinding

Specific Cutting energy $= \dfrac{\text{Power}}{\text{MRR}}$ $\quad F_c V_c$

G-ratio $= \dfrac{\text{Volume of w/p material removed}}{\text{Volume of wheel wear}}$

$= \dfrac{l \times w \times d \ (\text{depth of cut})}{\frac{\pi}{4}(D_o^2 - D_i^2) \times L}$

$\text{Time/cut} = \dfrac{L}{FN}$



$L = AP + CAP + l + OR$

$CAP = \sqrt{d(D-d)}$ -

$CAP = \frac{1}{2}\left(D - \sqrt{D^2 + W^2}\right)$

Wheel designation

① type of abrasive ② grain num ③ grade ④ structure ⑤ Type of Bond

$\boxed{52 - A - 220 - H - 7 - V - 36}$

① Type of abrasives

A - Al_2O_3 - Aluminium oxide

B = B_4C - Boron carbide

C - SiC - Silicon carbide

D - Diamond

WC = Tungsten carbide

② Grit size / grain num

10-24 - Roughing

30-120 - Semi-finishing

120-220 - finishing

240-600 - Super finishing

③ Grade / Hardness

A-H - Soft wheel

I-P - medium wheel

Q-Z = Hard wheel

④ Structure

0 to 7 close / dense structure

8 to 16 - open structure

⑤ Types of Bonding material

V - Vetrified bond

R - Rubber bond

B - Borozone / Resonord

M - metallic

S - Shellac